CYCLONES, STORM SURGE AND COASTAL PREPAREDNESS

COMPOUND CYCLONE HAZARD MAP → CYCLOGENESIS BOUNDARY → COASTAL WATER FIREWALL → IMD INFORMATION RAIL → COLOUR-STAGE MATRIX → WARNING-TO-EVACUATION CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◇ SUBORDINATE EXTRA ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

00 →

STAGE 00 COMPOUND CYCLONE HAZARD MAP

COMPOUND CYCLONE HAZARD MAP | STRUCTURE DEBRIS POWER TELECOM | RIVERINE FLASH PLUVIAL URBAN FLOOD | STORM SURGE + WAVES | COASTAL INUNDATION EROSION | TRANSPORT HEALTH WATER AND LIVELIHOODS

WIND → structure debris power telecom

RAIN → riverine flash pluvial urban flood → STORM SURGE + WAVES → coastal inundation erosion → CASCADE → transport health water and livelihoods

HAZARD / RISK / RESILIENCE CONCEPT

- WIND → structure debris power telecom
- RAIN → riverine flash pluvial urban flood

INSTITUTIONAL OR COMMUNITY RESPONSE

- STORM SURGE + WAVES → coastal inundation erosion
- CASCADE → transport health water and livelihoods

CAPACITY / STATUS / EVIDENCE LIMIT

- WIND → structure debris power telecom
- RAIN → riverine flash pluvial urban flood

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORM SURGE + WAVES → coastal inundation erosion.

01 →

STAGE 01 CYCLOGENESIS BOUNDARY

CYCLOGENESIS BOUNDARY | WARM WATER + INSTABILITY + MOISTURE | CORIOLIS + DISTURBANCE + LIMITED SHEAR | CYCLONIC SYSTEM | DM FOCUS | EXPOSURE WARNING ACTION CONTINUITY

WARM WATER + INSTABILITY + MOISTURE → CORIOLIS + DISTURBANCE + LIMITED SHEAR → CYCLONIC SYSTEM → DM FOCUS → exposure warning action continuity

HAZARD / ACTOR / INSTRUMENT

- WARM WATER + INSTABILITY + MOISTURE
- CORIOLIS + DISTURBANCE + LIMITED SHEAR

DECISIVE RISK OR MANDATE DISTINCTION

- CYCLONIC SYSTEM
- DM FOCUS → exposure warning action continuity

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- WARM WATER + INSTABILITY + MOISTURE
- CORIOLIS + DISTURBANCE + LIMITED SHEAR

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CYCLONIC SYSTEM; DM FOCUS → exposure warning action continuity.

02 →

STAGE 02 COASTAL WATER FIREWALL

COASTAL WATER FIREWALL | STORM SURGE | CYCLONE WIND AND PRESSURE | ASTRONOMICAL GRAVITATIONAL CYCLE | SUDDEN WATER DISPLACEMENT | TOTAL WATER LEVEL | INTERACTING BUT DISTINCT COMPONENTS

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
STORM SURGE → cyclone wind and pressure	TSUNAMI → sudden water displacement	STORM SURGE → cyclone wind and pressure
TIDE → astronomical gravitational cycle	TOTAL WATER LEVEL → interacting but distinct components	TIDE → astronomical gravitational cycle

HAZARD / ACTOR / INSTRUMENT

- STORM SURGE → cyclone wind and pressure
- TIDE → astronomical gravitational cycle

DECISIVE RISK OR MANDATE DISTINCTION

- TSUNAMI → sudden water displacement
- TOTAL WATER LEVEL → interacting but distinct components

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- STORM SURGE → cyclone wind and pressure
- TIDE → astronomical gravitational cycle

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TOTAL WATER LEVEL → interacting but distinct components.

03 →

STAGE 03 IMD INFORMATION RAIL

IMD INFORMATION RAIL | 1 OBSERVE | 2 CLASSIFY | FORECAST TRACK AND INTENSITY | 3 ISSUE IMPACT AND HAZARD INFORMATION | 4 UPDATE AS UNCERTAINTY CHANGES

HAZARD / RISK / RESILIENCE CONCEPT

- 1 OBSERVE / ANALYSE
- 2 CLASSIFY / FORECAST TRACK AND INTENSITY

INSTITUTIONAL OR COMMUNITY RESPONSE

- 3 ISSUE IMPACT AND HAZARD INFORMATION
- 4 UPDATE AS UNCERTAINTY CHANGES

CAPACITY / STATUS / EVIDENCE LIMIT

- 1 OBSERVE / ANALYSE
- 2 CLASSIFY / FORECAST TRACK AND INTENSITY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 ISSUE IMPACT AND HAZARD INFORMATION; 4 UPDATE AS UNCERTAINTY CHANGES.

04 →

STAGE 04 COLOUR-STAGE MATRIX

COLOUR-STAGE MATRIX | ACTION LEVEL | BULLETIN STAGE | LEAD-TIME SEQUENCE | REQUIRE LOCAL ACTION PROTOCOL

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
GREEN / YELLOW / ORANGE / RED → ACTION LEVEL	COLOUR != LEAD-TIME SEQUENCE	GREEN / YELLOW / ORANGE / RED → ACTION LEVEL
WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE	BOTH → require local action protocol	WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE

HAZARD / ACTOR / INSTRUMENT

- GREEN / YELLOW / ORANGE / RED → ACTION LEVEL
- WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE

DECISIVE RISK OR MANDATE DISTINCTION

- COLOUR != LEAD-TIME SEQUENCE
- BOTH → require local action protocol

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- GREEN / YELLOW / ORANGE / RED → ACTION LEVEL
- WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATCH / ALERT / WARNING / POST-LANDFALL → BULLETIN STAGE.

05 →

STAGE 05 WARNING-TO-EVACUATION CHAIN

WARNING-TO-EVACUATION CHAIN | OFFICIAL FORECAST | AREA DECISION | TRANSPORT + ROUTE + LIVESTOCK | ASSET PLAN | ACCESSIBLE SHELTER + REGISTRATION | ALL CLEAR | COMPETENT AUTHORITY

OFFICIAL FORECAST → AREA DECISION → TRANSPORT + ROUTE + LIVESTOCK / ASSET PLAN → ACCESSIBLE SHELTER + REGISTRATION → ALL CLEAR / RETURN → COMPETENT AUTHORITY

RISK CONDITION / INPUT

- OFFICIAL FORECAST → AREA DECISION
- TRANSPORT + ROUTE + LIVESTOCK / ASSET PLAN

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- ACCESSIBLE SHELTER + REGISTRATION
- ALL CLEAR / RETURN → COMPETENT AUTHORITY

LAST-MILE / INCLUSION / OUTCOME LIMIT

- OFFICIAL FORECAST → AREA DECISION
- TRANSPORT + ROUTE + LIVESTOCK / ASSET PLAN

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT + ROUTE + LIVESTOCK / ASSET PLAN.

06 →

STAGE 06 SHELTER READINESS TEST

SHELTER READINESS TEST | SAFE SITE AND ACCESS? | WATER SANITATION POWER ACCESSIBILITY? | TRAINED MANAGEMENT AND MAINTENANCE? | ASSET EXISTS BUT READINESS UNPROVEN

SAFE SITE AND ACCESS?

WATER SANITATION POWER ACCESSIBILITY? → TRAINED MANAGEMENT AND MAINTENANCE? → NO → ASSET EXISTS BUT READINESS UNPROVEN

HAZARD / RISK / RESILIENCE CONCEPT

- SAFE SITE AND ACCESS?
- WATER SANITATION POWER ACCESSIBILITY?

INSTITUTIONAL OR COMMUNITY RESPONSE

- TRAINED MANAGEMENT AND MAINTENANCE?
- NO → ASSET EXISTS BUT READINESS UNPROVEN

CAPACITY / STATUS / EVIDENCE LIMIT

- SAFE SITE AND ACCESS?
- WATER SANITATION POWER ACCESSIBILITY?

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAINED MANAGEMENT AND MAINTENANCE?; NO → ASSET EXISTS BUT READINESS UNPROVEN.

07 →

STAGE 07 CRITICAL-SERVICE WEB

CRITICAL-SERVICE WEB | SUPPLY CHAIN | EOC / HOSPITAL / RESCUE | POWER / WATER / TELECOM

EOC / HOSPITAL / RESCUE

POWER / WATER / TELECOM → ROAD / PORT / SUPPLY CHAIN → REDUNDANCY → ASSESS → RESTORE

HAZARD / RISK / RESILIENCE CONCEPT

- EOC / HOSPITAL / RESCUE
- POWER / WATER / TELECOM

INSTITUTIONAL OR COMMUNITY RESPONSE

- ROAD / PORT / SUPPLY CHAIN
- REDUNDANCY → ASSESS → RESTORE

CAPACITY / STATUS / EVIDENCE LIMIT

- EOC / HOSPITAL / RESCUE
- POWER / WATER / TELECOM

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ROAD / PORT / SUPPLY CHAIN; REDUNDANCY → ASSESS → RESTORE.

08 →

STAGE 08 HOUSING-TO-ECOSYSTEM PORTFOLIO

HOUSING-TO-ECOSYSTEM PORTFOLIO | SITING CODE COMPLIANCE MAINTENANCE | LIFELINE HARDENING AND REDUNDANCY | MANGROVE DUNE WETLAND PROTECTION | LAYERED DEFENCE WITH RESIDUAL RISK

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
HOUSING → siting code compliance maintenance	ECOSYSTEM → mangrove dune wetland protection	HOUSING → siting code compliance maintenance
INFRASTRUCTURE → lifeline hardening and redundancy	RULE → layered defence with residual risk	INFRASTRUCTURE → lifeline hardening and redundancy

HAZARD / ACTOR / INSTRUMENT

- HOUSING → siting code compliance maintenance
- INFRASTRUCTURE → lifeline hardening and redundancy

DECISIVE RISK OR MANDATE DISTINCTION

- ECOSYSTEM → mangrove dune wetland protection
- RULE → layered defence with residual risk

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- HOUSING → siting code compliance maintenance
- INFRASTRUCTURE → lifeline hardening and redundancy

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ECOSYSTEM → mangrove dune wetland protection; RULE → layered defence with residual risk.

09 →

STAGE 09 NCRMP STATUS LADDER

NCRMP STATUS LADDER | PROJECT APPROVAL | ASSET CREATION | PHASE COMPLETION | LOCAL HAND-OFF | OPERATION + MAINTENANCE + DRILLS | CURRENT READINESS | NEEDS SEPARATE EVIDENCE

PROJECT APPROVAL → ASSET CREATION

PHASE COMPLETION → STATE / LOCAL HAND-OFF → OPERATION + MAINTENANCE + DRILLS → CURRENT READINESS → needs separate evidence

RISK CONDITION / INPUT

- PROJECT APPROVAL → ASSET CREATION
- PHASE COMPLETION → STATE / LOCAL HAND-OFF

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- OPERATION + MAINTENANCE + DRILLS
- CURRENT READINESS → needs separate evidence

LAST-MILE / INCLUSION / OUTCOME LIMIT

- MUST REMEMBER: Cyclone disaster risk joins wind, storm surge, heavy rainfall, waves,..
- PROJECT APPROVAL → ASSET CREATION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Cyclone disaster risk joins wind, storm surge, heavy rainfall, waves,.

10 →

STAGE 10 LIVELIHOOD RECOVERY LOOP

LIVELIHOOD RECOVERY LOOP | BOAT GEAR LIVESTOCK AND MARKET ACTION | RELIEF + SOCIAL PROTECTION | SAFER CONDITIONS | BUILD BACK BETTER | REDUCE FUTURE EXPOSURE | CLOSE DISTINCTION | CYCLONE CATEGORY IS NOT STORM-SURGE HEIGHT; FORECAST LANDFALL IS

WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION

IMPACT → RELIEF + SOCIAL PROTECTION → REPAIR / REOPEN → SAFER CONDITIONS → BUILD BACK BETTER → REDUCE FUTURE EXPOSURE

HAZARD / RISK / RESILIENCE CONCEPT

- WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION
- IMPACT → RELIEF + SOCIAL PROTECTION

INSTITUTIONAL OR COMMUNITY RESPONSE

- REPAIR / REOPEN → SAFER CONDITIONS
- BUILD BACK BETTER → REDUCE FUTURE EXPOSURE

CAPACITY / STATUS / EVIDENCE LIMIT

- CLOSE DISTINCTION: Cyclone category is not storm-surge height; forecast landfall is not...
- WARNING → BOAT GEAR LIVESTOCK AND MARKET ACTION

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Cyclone category is not storm-surge height; forecast landfall is not.

11 →

STAGE 11 CYCLONE ANSWER SPINE

CYCLONE ANSWER SPINE | DEFINE COMPOUND HAZARD | SEPARATE SURGE TIDE TSUNAMI | MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE | CONNECT EVACUATION SHELTER LIFELINES HOUSING AND ECOSYSTEMS | ADD LIVELIHOODS BBB MAINTENANCE AND OUTCOME QUALIFICATION | EVIDENCE LIMIT | USE IMD TERMINOLOGY AND TIMESTAMPED

DEFINE COMPOUND HAZARD → SEPARATE SURGE TIDE TSUNAMI → MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE → CONNECT EVACUATION SHELTER LIFELINES HOUSING AND ECOSYSTEMS → ADD LIVELIHOODS BBB MAINTENANCE AND OUTCOME QUALIFICATION → EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use IMD terminology and timestamped

HAZARD / RISK / RESILIENCE CONCEPT

- DEFINE COMPOUND HAZARD → SEPARATE SURGE TIDE TSUNAMI
- MAP IMD CLASSIFICATION FORECAST COLOUR AND BULLETIN STAGE

INSTITUTIONAL OR COMMUNITY RESPONSE

- CONNECT EVACUATION SHELTER LIFELINES HOUSING AND ECOSYSTEMS
- ADD LIVELIHOODS BBB MAINTENANCE AND OUTCOME QUALIFICATION

CAPACITY / STATUS / EVIDENCE LIMIT

- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use IMD terminology and timestamped...
- DEFINE COMPOUND HAZARD → SEPARATE SURGE TIDE TSUNAMI

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use IMD terminology and timestamped...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Use IMD terminology and timestamped.

E →

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT | MULTI-SECTORAL PREMISE, EXPLICITLY STATED | NDMA'S 2008 CYCLONE | NCRMP'S INSTITUTIONAL STRUCTURE | NDMA (UNDER MHA) IMPLEMENTS IN | ASYMMETRIC COASTAL EXPOSURE, PRECISELY QUANTIFIED IN THE SOURCE | 1994 ODISHA CYCLONE AS A CASCADING-HAZARD CASE | AIRCRAFT PROBING OF CYCLONE (APC) AS AN UNDER-INSTITUTIONALISED

OPTIONAL RISK / RESILIENCE DEPTH

- Multi-sectoral premise, explicitly stated : NDMA's 2008 cyclone
- NCRMP's institutional structure : NDMA (under MHA) implements in
- Asymmetric coastal exposure, precisely quantified in the source :
- CAUTION: 1994 Odisha cyclone as a cascading-hazard case : cyclone

MANDATE, WARNING OR OUTCOME LIMIT

- CAUTION: Aircraft Probing of Cyclone (APC) as an under-institutionalised
- CAUTION: VisionIAS names "a lack of a fully automated and state-of-the-art
- CAUTION: VisionIAS's "lack of grass root level participation in disaster
- CAUTION: Category-based investment prioritisation is itself a trade-off.

COMPARATIVE HAZARD / GOVERNANCE NUANCE

- CAUTION: Saline-embankment/bio-shield maintenance versus new-construction
- NOT: The "NCRMP exists, therefore coastal States are protected"
- CAUTION: Network fragmentation across agencies : the explicit call for
- NCRMP's blended financing (NDMA/MHA-led, State-coordinated,

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels